

Dev Track

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DevTrack: Enhancing Transparency and Accountability in Kathmandu's Development Projects

Executive summary:

DevTrack is an innovative web application designed to bridge the communication gap by bringing transparency and accountability to government projects in Kathmandu Metropolitan City (KMC). It provides comprehensive details on project budgets , quality , schedules , authorities and usage , ensuring the general public has easy access to vital data . This transparency builds trust between the KMC government and its citizens , reducing corruption and mismanagement while promoting a culture of openness . Additionally DevTrack's real time monitoring feature allows citizens to track project progress , hold authorities accountable for delays , and facilitate better communication and engagement . Moreover , DevTrack includes a public forum where users can publicly comment on development projects and engage in discussions . This forum empowers citizens to share their views and feedback directly with authorities and other stakeholders . Users will also have access to detailed information about development initiatives specifically targeted at them , enabling them to understand and influence the projects that impact their communities directly . This dual functionality ensures that DevTrack not only provides transparency but also promotes active citizen participation in the development.

Purpose: DevTrack is a web application designed to improve transparency and accountability in government projects in Kathmandu Metropolitan City (KMC).

Key Features:

- **Real-Time Monitoring:** Track project progress, hold authorities accountable, and facilitate better communication.
- **Public Forum:** Allows citizens to comment on and discuss development projects.
- **Comprehensive Data:** Provides details on budgets, quality, schedules, and authorities.
- **User Customization:** Personalized dashboards based on user interests.

Background Information:

- **Context:** Kathmandu Valley is experiencing rapid urbanization with numerous development projects.
- **Problem:** Lack of transparency and public awareness leading to delays, budget overruns, and lack of accountability.
- **Solution:** DevTrack provides a centralized platform for monitoring and reporting on development projects.

Missions:

Goals:

- Enhance community engagement and communication.
- Improve financial accountability and transparency.
- Ensure accessible and clear information on development projects.
- Create public forum such that all the important opinions are given to the government.
- Ensure opinions of experts to be prioritized to give valuable insights to the government such that development projects have constructive feedback.
- Prove good governance and create a public forum such that international bodies have a platform to ensure KMC is encouraging public participation.
- Further, KMC can be a part of OGP(Open Government Partnership)

Approach: Utilize a public forum to channelize citizen concerns and bridge communication gaps.

Project Implementation:

Public Surveys:

- Conducted surveys indicating a high demand for a project tracker.

Data Collection and Categorization:

- Scrape data from governmental websites and outreach contacts.
- Categorize data into eight categories: Infrastructure, Education, Healthcare, Economy, Environment, Policy, Urban Development, and Citizen Benefits.

User Interaction and Customization:

- Allow users to create profiles and customize dashboards.
- Features include project timelines, interactive maps, and visualizations.

Technical Implementation:

- Technologies: Next.js for server-side rendering, Supabase for backend, Node.js.
- Features: Mapping, visualization, user feedback systems for reporting project progress.

Tangible Outcome of this project:

By completing the DevTrack project, we aim to deliver a centralized, user-friendly web platform that provides comprehensive information on governmental projects in Kathmandu Valley. Our website will include a detailed database, customizable user dashboards, feedback mechanisms, and analytical tools to ensure transparency and accountability. The platform will enable real-time monitoring, facilitating public engagement and participation. Ultimately, DevTrack seeks to enhance public services, improve communication gap, and foster a culture of transparency and accountability, leading to long-term systemic changes in project management.

Sustainability:

Our project and its impact is self sustainable. As the platform gains traction, it is anticipated that governmental agencies will adopt DevTrack as their primary means of disseminating project information to the public. This will not only foster greater public awareness and engagement but also enhance trust in government operations. Additionally, the platform's scalable and adaptable design ensures that it can evolve with technological advancements and changing user needs, securing its place as a vital resource for tracking and managing government projects.

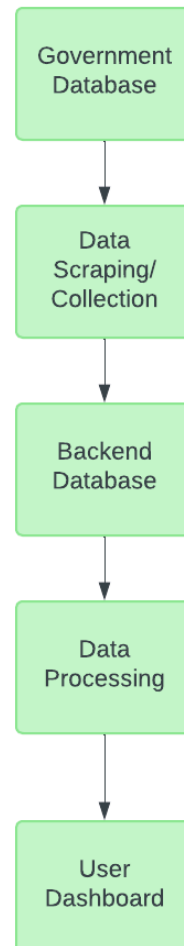
Technical Architecture and User Flow

1. Structure Diagram: Data Flow from Government Databases to the User Dashboard

Outline:

1. **Government Databases**
 - Source of project data.
 - Data includes project budgets, timelines, status, authorities involved, etc.
2. **Data Scraping/Collection**
 - Automated scripts or manual collection methods.
 - Data categorization (Infrastructure, Education, Healthcare, etc.)
3. **Backend Database (Supabase)**
 - Secure storage of collected data.
 - Data normalization and indexing.
4. **Data Processing**
 - Data analytics.
 - Generating summaries and insights.
5. **User Dashboard**
 - User-specific customization.
 - Display of project timelines, interactive maps, visualizations, and feedback mechanisms.

Flowchart Representation

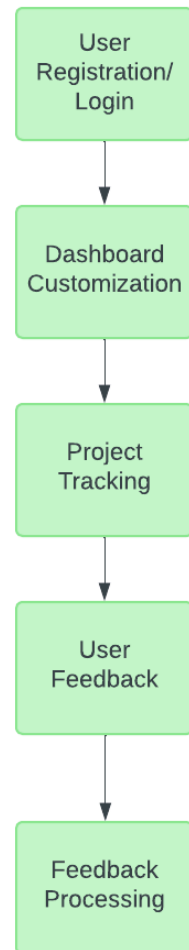


2. User Interaction Flowchart: User Journey from Creating a Profile to Providing Feedback

Outline:

1. **User Registration/Login**
 - Create a profile or log in.
 - Specify interests (Infrastructure, Healthcare, etc.)
2. **User Dashboard Customization**
 - Dashboard populated based on user interests.
 - Display of relevant projects, timelines, and maps.
3. **Project Tracking**
 - Users track project progress.
 - Real-time updates and notifications.
4. **User Feedback**
 - Users provide feedback on project status.
 - Engage in discussions on the public forum.
5. **Feedback Processing**
 - Feedback reviewed by authorities.
 - Updates and responses posted on the platform

Flowchart Representation



Literature Review on DevTrack's Feasibility

The literature on governance, transparency, and accountability in development projects underscores the importance of open access to information and citizen engagement in enhancing the effectiveness and efficiency of these initiatives. This review synthesizes key findings from scholarly works, reports, and case studies, with a particular focus on the context of Nepal and the Kathmandu Valley.

Governance and Transparency

Global Perspectives

Governance and transparency are foundational to sustainable development, as highlighted by numerous scholars and organizations. According to the World Bank (2016), transparency in government projects enhances accountability, reduces corruption, and improves public trust. Open access to information allows citizens to monitor project progress, which is crucial for ensuring that projects meet their intended outcomes.

Nepalese Context

In the context of Nepal, the challenges and opportunities for enhancing governance and transparency in development projects have been extensively studied. Kharel (2019) emphasizes that the lack of transparency in Nepalese development projects often leads to inefficiencies and corruption. He argues that a centralized platform providing real-time updates on project status could significantly mitigate these issues.

Importance of Citizen Engagement

Global Perspectives

Citizen engagement is another critical aspect of effective governance. Fung (2006) asserts that involving citizens in the decision-making process leads to more democratic governance and better project outcomes. Citizens' direct feedback can provide valuable insights and help ensure that projects align with local needs and priorities.

Nepalese Context

Stone (1989) highlights the importance of community participation in Nepal's development projects. He notes that engaging local communities not only enhances the relevance and sustainability of projects but also fosters a sense of ownership among residents. However, there is often a gap between policy and practice, with limited mechanisms for meaningful citizen participation.

Financial Accountability

Global Perspectives

Financial accountability is crucial for the successful implementation of development projects. According to Agwor (2017), clear and accessible financial reporting helps prevent mismanagement and corruption. It also builds trust among stakeholders, which is essential for securing ongoing support and funding.

Nepalese Context

In Nepal, financial transparency remains a significant challenge. A study by Kharel (2019) reveals that many development projects suffer from budget overruns and delays due to poor financial management. He advocates for the adoption of international best practices in financial reporting and the establishment of robust monitoring mechanisms.

Technological Solutions for Transparency

Global Perspectives

Advances in technology have provided new tools for enhancing transparency and accountability in development projects. Platforms that utilize data analytics and interactive mapping can offer real-time insights into project progress and financial status. According to Kassen (2017), such technological solutions can transform governance by making information more accessible and engaging for the public.

Nepalese Context

In Nepal, the use of technology for enhancing governance is still in its nascent stages. However, initiatives like the Nepal e-Government Interoperability Framework (NeGIF) demonstrate the potential for leveraging digital tools to improve transparency and accountability. A study by Shakya (2018) underscores the need for more comprehensive and user-friendly platforms that provide detailed information on development projects.

Case Studies

Global Examples

Successful examples of transparency platforms include the World Bank's Projects & Operations portal. These platforms provide detailed information on project objectives, funding, progress, and outcomes, demonstrating how transparency can be effectively implemented on a large scale World Bank (2016).

Nepalese Examples

In Nepal, the Public Procurement Monitoring Office (PPMO) has developed an online system for monitoring public procurement. While this is a positive step, Shakya (2018) notes that there is still a need for more comprehensive platforms that cover all aspects of development projects, including those outside the scope of public procurement.

In conclusion, The literature reviewed highlights the critical role of governance, transparency, and citizen engagement in the successful implementation of development projects. While there are significant challenges, particularly in the context of Nepal, there are also promising opportunities to leverage technology and best practices to enhance accountability and public trust. The proposed DevTrack platform aligns well with these findings, aiming to provide a centralized, transparent, and user-friendly resource for tracking development projects in the Kathmandu Valley.

Fields we are trying to cover in our project:

After we have a detailed list of projects our next step is to categorize the projects. We will be sorting this manually for now. The projects will be classified based on the following sectors:

Infrastructure Development : Roads and highways, Bridges and tunnels, Airports, Public transportation system, Water supply and sanitation

Educational Development : Construction and renovation of schools and universities, Curriculum development, Scholarship program

Healthcare Development : Building and equipping hospitals and clinics, Public health campaigns (vaccination , disease prevention), Training healthcare workers

Economic Development : Small business and entrepreneurship support, Vocational training and job creation programs, Microfinance initiatives, Industrial park development

Environmental Development : Reforestation and afforestation projects, Conservation of natural resources, Renewable energy projects (solar, wind, hydro)

Policy Development : Formulation of economic policies (eg: tax policy , labour laws), Environmental regulation policies, Healthcare policies etc

Urban Development : Housing projects, Urban renewal and slum upgrading, Public parks and recreational facilities, Smart city initiatives

Citizen Benefits : for instance filling application for allowances and things

Design Prototype Images:

Figure 1: Project sorting feature with mapline

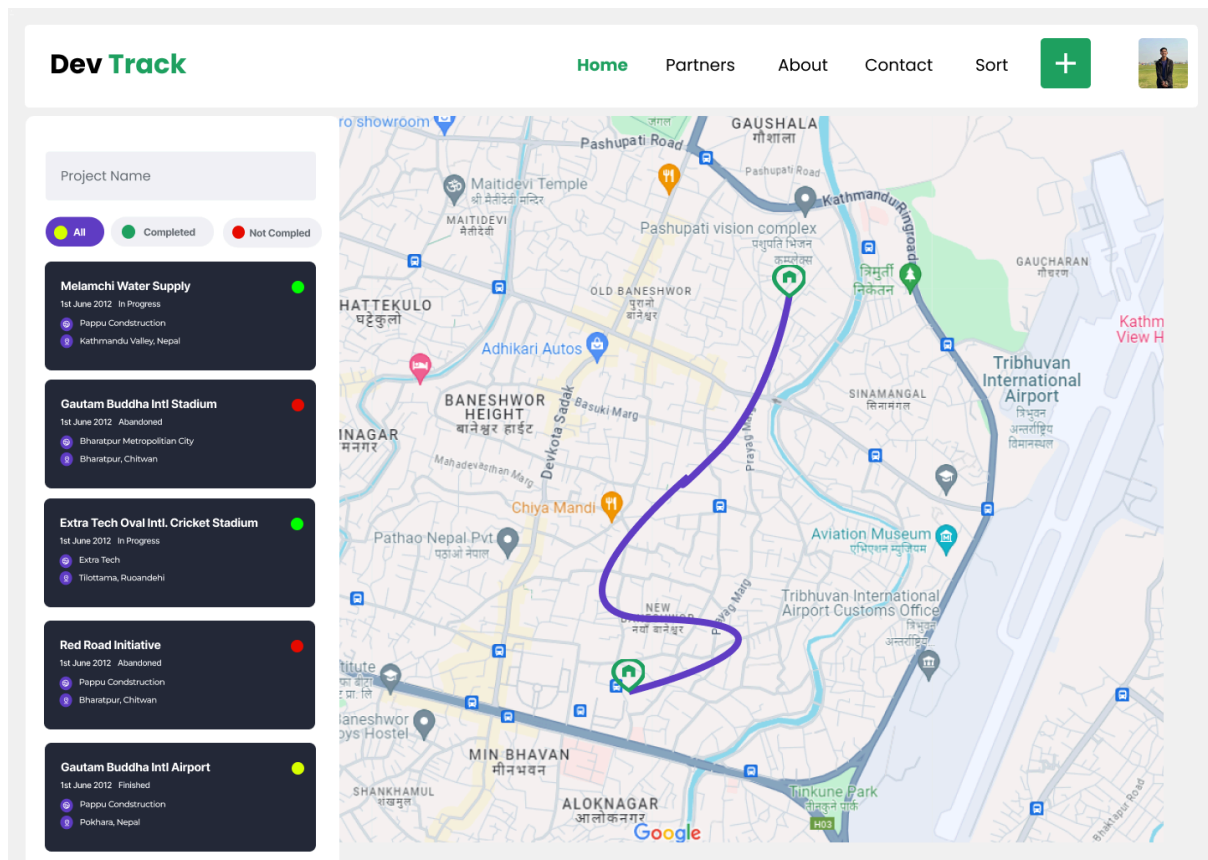
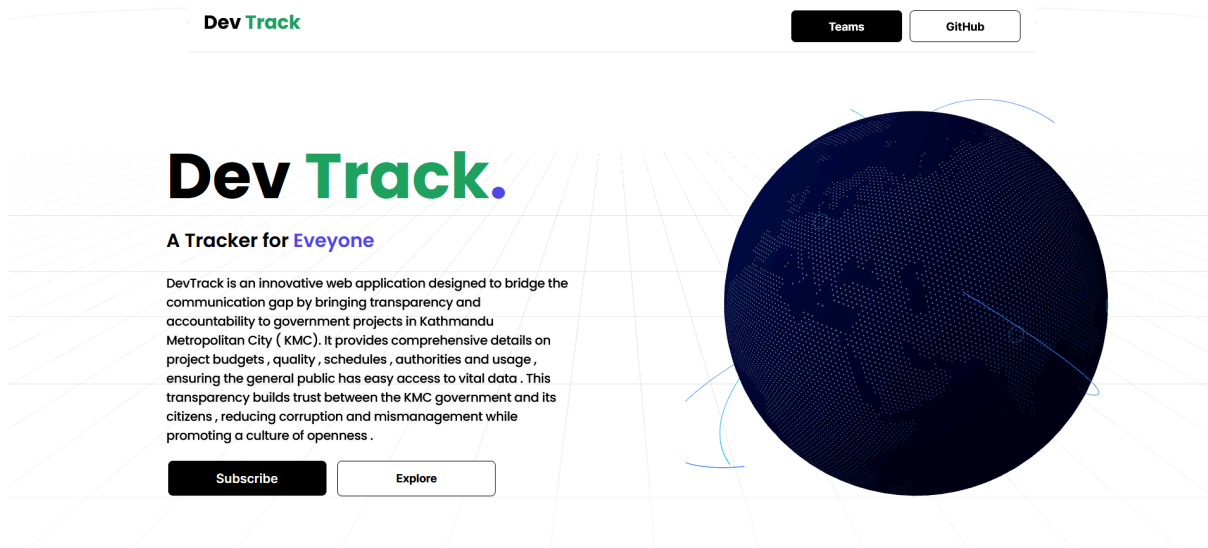


Figure 2: Landing page of our website



Domain: www.devtrack.org

References

- Kharel, S. (2019). Transparency and accountability in the Nepalese corporate sector: A critical assessment. *Kharel, S., Magar, S., Chaurasiya, N., Maharjan, S. & Rijal, CP (2019). Transparency and Accountability in the Nepalese Corporate Sector: A Critical Assessment. Quest Journal of Management and Social Sciences: Corporate Governance Edition, 1(1), 1-25.*
- Fung, A. (2006). Varieties of Participation in Complex Governance. *Public Administration Review*, 66(s1), 66-75. <https://doi.org/10.1111/j.1540-6210.2006.00667.x>
- Shakya, S. (2018). E-Governance in Nepal: Progress, Challenges and Possibilities.
- Kassen, M. (2017). Open Data and e-Government – Related or Competing Ecosystems: A Conceptual Analysis. *Government Information Quarterly*, 34(1), 45-53. <https://doi.org/10.1016/j.giq.2016.08.001>
- Stone, L. (1989). Cultural crossroads of community participation in development: a case from Nepal. *Human organization*, 48(3), 206-213.
- Agwor, T. C., & Akani, F. N. (2017). Financial accountability and performance of local governments in rivers state, Nigeria. *International Journal of Economics, Commerce and Management*, 5(10), 620-635.
- World Bank Group. (2016). *World development report 2016: Digital dividends*. World Bank Publications.

About Us:

We are a team of six student researchers, guided by a mentor and a peer mentor, working in one of the six projects in Incubate Nepal 2024. Incubate Nepal is an 8-week virtual program that connects some of the most talented students from all over Nepal to collaborate on open-ended projects.

Shubham Jha - Mentor

Shubham is a junior at Franklin and Marshall College, where he is pursuing a dual degree in Economics and Mathematics. His academic journey is complemented by a profound commitment to social impact, particularly in building grassroots social enterprises and engaging with marginalized communities to address their challenges. Shubham was the first student in the history of F&M college to be nominated for the Davis Project for Peace Fellowship in his freshman year and he established a program to help Nepalese Waste Entrepreneurs. Last spring, Shubham interned at the Office of Critical Investments, Commonwealth of Pennsylvania where he was a part of the team to strategically allocate \$20bn federal funding to workforce development and energy security projects. Shubham is passionate about building efficient governing systems and has gained profound experience working with Nepalese and the United States government.

Prashim Timsina - Peer Mentor

Prashim is a high school graduate from Trinity Int'l College with a major in computer science. He is regarded as a tech geek, although he doesn't consider himself one. He is involved in various Technopreneur activities bringing solutions to existing problems in the society through his tech. Other than that, he enjoys playing guitar as a coping mechanism and has hobbies that include photography, cinematography, traveling, and a lifelong dream of summiting Mount Everest. As a great peer mentor, he takes pride in fostering a productive working environment for the mentees.

Members

Sugam Parajuli

Sugam is a versatile member of Team DevTrack, bringing a wide range of skills and interests to the project. He excels in math and has a keen interest in psychology, computer science, and AI. Currently running his own computer shop in Pokhara, Sugam is adept at report writing, photography, and documentation, and has a passion for video-making.

Preeti Pantha

Preeti is a socio-economic enthusiast with a deep passion for developmental economics. She enjoys crocheting, cycling, and Bollywood music. Known for her sense of humor, Preeti is excited to collaborate with the team to create something impactful and memorable.

Tushar Shah

Tushar is an enthusiastic thinker with a keen interest in socio-economic issues. Tushar believes that "everything is achievable" and aims to be on the cover of Forbes one day.

Sampada Koirala

Sampada is an economics enthusiast who loves capturing life's bloopers. When she's not engaged in these pursuits, you can find her strumming her guitar and pretending to be a rockstar, though her neighbors might disagree! Despite being introverted, her friends know her as a funny person. Sampada is excited to learn alongside her team members and mentors.

Sushil Bhattarai

Sushil Bhattarai is a passionate Computer Science enthusiast dedicated to leveraging technology to create positive social impact. Actively involved in various community organizations, Sushil has been using his expertise to bridge the gap between communities and modern technological advancements. His work is driven by a vision to create a world where technology empowers people and provides equal opportunities for all. Sushil's commitment to community service and innovative tech solutions reflects his belief in technology as a powerful tool for societal betterment.

Kristina Khanal

Kristina is a talented computer science enthusiast with a knack for designing impressive websites. Known for her love of conversation, Kristina generates brilliant ideas while engaging in lively discussions. She actively participates in national hackathons as a Tech Girl, showcasing her skills and creativity. Outside of her tech pursuits, Kristina enjoys brainstorming innovative solutions and connecting with like-minded individuals.